DNS Guard AI

Behavioral DNS threat detection with Splunk MLTK

Riccardo Alesci

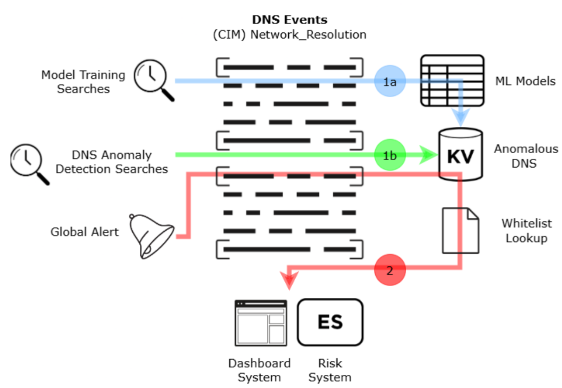
1 Problem Statement

The **Domain Name System (DNS)** is a fundamental pillar of modern network communication, responsible for resolving hostnames to IP addresses. Despite its critical role, **DNS** often remains **under-monitored** within enterprise security strategies, making it an attractive vector for threat actors. Advanced adversaries increasingly leverage DNS to establish **command-and-control (C2)** channels, exfiltrate **sensitive information**, and perform **stealth reconnaissance activities—all** while blending within legitimate traffic patterns. These attacks frequently exploit the inherent trust and ubiquity of **DNS**, using low-volume, protocol-compliant queries that bypass signature-based intrusion detection systems. The subtlety and evasiveness of **DNS-based threats** demand a behavioral and data-driven approach to detection.

DNS Guard AI addresses this security gap by applying unsupervised machine learning and statistical analysis to **DNS telemetry** within the Splunk platform. By modeling baseline DNS behavior and detecting deviations indicative of malicious activity, **DNS Guard AI** provides real-time visibility into anomalous patterns that traditional tools fail to uncover—offering a scalable and adaptive defense mechanism for enterprise environments.

2 Solution Overview

The architecture illustrates how DNS Guard AI processes DNS events mapped to the **Network_Resolution** data model in Splunk. **Model training** searches extract features from historical DNS traffic to train machine learning models via the **MLTK** (1a). In parallel, **anomaly detection** searches continuously scan incoming DNS data to identify suspicious behavior such as **exfiltration**, **tunneling**, or **domain shadowing** (1b). Detected anomalies are stored in a **KV Store collection** and compared against a **whitelist** to suppress false positives. Validated anomalies are then sent to two systems: the **dashboard interface** for visual monitoring, and **Splunk Enterprise Security** (ES) for risk scoring and alert generation (2). This design ensures scalable, real-time DNS threat detection tightly integrated with Splunk's security ecosystem.



3 Machine Learning Approach

The detection engine of **DNS Guard** AI integrates multiple unsupervised learning algorithms to identify deviations from normal DNS behavior across diverse threat dimensions:

- **Density Function Algorithm**
  - **Beaconing**: Detects regular, periodic DNS queries at consistent intervals—a hallmark of malware communicating with command and control servers. Analyzes consistency of time gaps between queries to the same domain
- **Anomaly Detection Algorithm**
  - **Record Type Anomalies**: Detects abnormal usage of specific DNS record types often associated with reconnaissance or data exfiltration. Identifies anomalies in the usage of TXT (data exfiltration), ANY (broad queries), HINFO (host info leakage), and AXFR (zone transfer attempts) records by host.
  - **C2 Tunneling Detection**: Identifies hosts making an unusually high number of DNS queries, which could indicate command and control communication or data exfiltration through DNS tunneling.
  - **Query Length Anomalies**: Detects unusually long DNS queries that may represent data exfiltration channels where sensitive information is encoded in the query itself. Identifies anomalies in query string length by host.

- **Domain Shadowing**: Identifies patterns where many unique subdomains are requested for a legitimate domain, which may indicate an attacker using compromised DNS accounts to create malicious subdomains. Measures distinct subdomain count by parent domain and identifies anomalies.

- **K Means Algorithm**
  - **Behavioral Clustering**: Groups hosts with similar abnormal DNS behavior, which can reveal coordinated attacks or infected host groups across the enterprise. Uses KMeans clustering on multiple DNS behavior features.

4 Installation and Configuration

Prerequisites

- Splunk Enterprise/Cloud 8.0+
- Splunk machine with at least **16 GB of RAM** is required for the app to run well

Install the following apps from Splunkbase:

- Splunk CIM App
- Splunk MLTK App
- Python for Scientific Computing

1. Install the App & Generate Test Data

- Clone the repo:

```
git clone https://github.com/aleeric/Splunk-DNS-Guard-AI.git
```

- Move the folder `Splunk-DNS-Guard-AI/` to `$SPLUNK_HOME/etc/apps/` and restart Splunk.
- Commands on Terminal

```
cd Synthetic-Data
python generate_dns_events.py
```

- New JSON file **dns_events.json** created

2. Import Synthetic Data into Splunk

Via Splunk Web:

- Go to **Settings** → **Add Data** → **Upload** files from my computer → **Select File** 'dns_events.json' → Click **Next**
- Select **'synthetic-data'** from **Source type** list → Click **Next** → **Host field value: 'dns-guard-simulator'** → Create a new index → **Index name: 'synthetic-data'** (Important!); leave the other fields as they are. → Click **Save** → Click **Review** → Click **Submit**

3. Map to Network_Resolution Model

- Go to **Apps** → **Manage Apps** → Search **'CIM'** → Click **Set up** → Search **"Network Resolution"** → Insert **'synthetic-data'** into **'Indexes allowlist'** field → Click **Save**

4. Increase Values on MLTK Settings

- Go to **Apps** → **Splunk Machine Learning Toolkit** → **Settings** → Click **DensityFunction** → Set **'max_groups'** value from 5000 to 500000 → Set **'max_inputs'** value from 100000 to 10000000 → Click **Save**

5. Verify Setup

- Open **DNS Guard AI** on Splunk Web Interface → Go to **setup** → Go to **'DNS Data Model'** page → Check **'Network Resolution Event Count'** value >0 → Click **Run Query** on all searches in this order: **C2 Tunnel Detection, Query Length Anomalies, Domain Shadowing, Record Type Anomalies (TXT, ANY, HINFO, AXFR). Behavioral Clustering, Beaconing (Upper), Beaconing (Lower)**. It's important to wait for each search to finish before starting the next.

5 Conclusion

DNS Guard AI gives security teams an advanced and scalable solution to detect DNS threats in real time. It combines machine learning, Splunk's search power, and a modular dashboard system to identify C2, exfiltration, and reconnaissance via DNS, filling a critical detection gap in enterprise defense.